LESSON PLAN

NDT (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Thrust augmentation and Thrust augmentation by water injection and water methanol injection system After burning operation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Thrust augmentation and Thrust augmentation by water injection and water methanol injection system After burning operation
5. Specific Objectives	20.1.2.1 Trainee will be able to understand and appreciate noise separation methods, thrust reversal mechanism, after burning fuel system, lubrication system, thrust augmentation devices, cooling and thrust calculations

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of PPT slides on Water Injection and Afterburning systems; Instructor supervision of student note-taking and Q&A; session.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of thrust augmentation in a jet engine?
(b) Name two methods used to increase thrust beyond standard limits.
(c) How does afterburning affect the temperature of the exhaust gases?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principles of Thrust Augmentation • Water Injection System Operation • Water Methanol Injection System • Afterburning Fuel System Components • Cooling and Thrust Calculations	L & Ds	PPT, IB	15 min
(b)	• Mechanism of Water Injection Cooling • Water Methanol Injection Process • Afterburner Ignition Sequence • Thrust Calculation Formulas • System Limitations and Safety	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Injection Systems • Impact on Engine Life • Noise Separation Methods • Thrust Reversal Mechanism • Lubrication System Interaction	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between water injection and water methanol injection regarding cooling efficiency.
(b) Describe the sequence of events during an afterburning operation.
(c) How do thrust augmentation devices impact the lubrication system requirements?

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of thrust augmentation discussed in this lesson apply to the noise suppression techniques covered in the next lesson plan (LP 11).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Noise suppression and its levels. Various methods of thrust reversal and principles of operation.	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min